

The HHI Standards Architecture: A Unified Governance Framework for Trustworthy AI Systems

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Abstract

This document, HHI-WP-2026-016, constitutes the capstone publication of the Hollow House Institute (HHI) Governance Canon: a sixteen-paper corpus of governance standards, protocols, and frameworks developed to address the structural deficiencies in global AI governance architecture. The HHI Standards Architecture is presented herein as the culminating synthesis of that corpus — a unified, layered, and interoperable governance framework designed to govern AI systems across their full operational lifecycle, from design-time risk architecture through runtime behavioral governance to longitudinal accountability and ecosystem-level trust.

The HHI Standards Architecture integrates fifteen foundational governance domains: trust ecosystem design, governance fragmentation response, cybersecurity convergence, execution-time enforcement, behavioral drift governance, sector-specific compliance, runtime provenance, longitudinal accountability, interoperability architecture, supply chain integrity,

emergent behavior governance, adversarial resilience, citation integrity, evidence-anchored writing, and governance observability. Each of these domains, developed as a standalone whitepaper within the Canon, is here reconstituted as a load-bearing architectural pillar. No pillar can be removed without inducing structural governance failure: the integrity of the whole depends upon the integrity of every constituent part.

The Architecture is organized into three hierarchical layers: Foundational Governance Infrastructure, which establishes the substrate of trust, provenance, interoperability, and supply chain integrity upon which all other governance depends; Operational Governance Systems, which govern AI behavior in production across execution, drift, observability, adversarial threats, and longitudinal time horizons; and Epistemic and Contextual Governance Standards, which govern the knowledge and regulatory contexts within which AI systems are developed, deployed, and evaluated. Together, these layers constitute the first unified, operationally deployable standards architecture for AI governance — built from first principles, grounded in evidence, and designed for enterprise, regulatory, and international standards body adoption.

The HHI Governance Canon represents an unprecedented contribution to the global governance commons. This capstone document equips governance professionals, enterprise architects, AI system operators, and international regulators with the architectural understanding, maturity model, implementation roadmap, standards alignment map, and organizational framework required to deploy the Architecture in full.

Executive Summary

The central question of contemporary AI governance is deceptively simple: *what does a complete governance system actually look like?* After more than a decade of framework proliferation, regulatory divergence, and well-intentioned but architecturally incomplete standards activity, the governance community remains without a rigorous, operationally coherent answer. The HHI Standards Architecture is that answer.

The Hollow House Institute was founded by Amy, a governance researcher and standards practitioner whose work identified a fundamental gap in the global AI governance landscape: not an absence of frameworks, but an absence of architecture. Individual frameworks — the NIST AI Risk Management Framework, the EU AI Act, ISO/IEC 42001:2023 — address specific governance problems with rigor and authority. What they do not do is compose. They do not form an integrated operational system capable of governing an AI system across its full lifecycle, across regulatory jurisdictions, and across the full spectrum of risk types that modern AI systems present. Amy founded HHI to build that architecture from first principles, and the Governance Canon is the result of that foundational commitment.

The sixteen-paper Canon addresses fifteen distinct governance domains across the full AI governance landscape. **HHI-WP-2026-001** established the foundational theory of trust ecosystem design, articulating how multi-stakeholder AI trust is architected rather than assumed. **HHI-WP-2026-002** analyzed global governance fragmentation and proposed a

coherent harmonization strategy for organizations operating across jurisdictional boundaries. **HHI-WP-2026-003** developed the convergence model for AI cybersecurity and AI governance, arguing that these disciplines must be treated as a unified domain. **HHI-WP-2026-004** introduced the Execution-Time Governance Framework, governing AI decisions at the moment of inference. **HHI-WP-2026-005** addressed behavioral drift — the gradual, often undetected departure of AI systems from their governance baselines. **HHI-WP-2026-006** delivered sector-specific governance kits for healthcare, financial services, critical infrastructure, and public administration. **HHI-WP-2026-007** established the Runtime Provenance Architecture, operationalizing W3C PROV-DM for AI systems. **HHI-WP-2026-008** introduced the Longitudinal Governance Standard, governing AI systems across multi-year operational horizons. **HHI-WP-2026-009** developed the Governance Interoperability Architecture, enabling composable compliance across frameworks. **HHI-WP-2026-010** established the Supply Chain Integrity Standard, extending governance to ML-BOM and third-party component provenance. **HHI-WP-2026-011** addressed emergent behavior — governance for AI capabilities not present or anticipated at design time. **HHI-WP-2026-012** delivered the Adversarial Governance Framework, aligned to MITRE ATLAS. **HHI-WP-2026-013** established the Citation Integrity Standard for AI governance documentation. **HHI-WP-2026-014** introduced the Evidence-Anchored Writing Standard for governance artifacts. **HHI-WP-2026-015** delivered the Governance Observability Architecture, operationalizing OpenTelemetry for AI governance telemetry.

These fifteen whitepapers compose into the HHI Standards Architecture through three architectural layers. **Layer 1: Foundational Governance Infrastructure** comprises Trust Ecosystem Design, Runtime Provenance Architecture, Supply Chain Integrity, and Governance Interoperability — the substrate without which no operational governance system can function. **Layer 2: Operational Governance Systems** comprises Execution-Time Governance, Behavioral Drift, Emergent Behavior, Adversarial Governance, Governance Observability, and Longitudinal Governance — the active governance mechanisms that operate AI systems in production. **Layer 3: Epistemic and Contextual Governance Standards** comprises Global Fragmentation Response, Harmonized Cyber Governance, Sector-Specific Governance, Citation Integrity, and Evidence-Anchored Writing — the contextual and knowledge standards that ensure governance is jurisdictionally coherent and epistemically sound.

This capstone document delivers the complete integration specification: the maturity model against which organizations can measure their governance posture; the standards alignment master map correlating all sixteen Canon domains to NIST AI RMF, ISO/IEC 42001, the EU AI Act, and NIST CSF 2.0; the four-phase enterprise implementation architecture; the organizational governance function model; the Canon publication and contribution framework; and the research agenda for the next generation of Canon expansion. The HHI Standards Architecture is not a final word. It is a rigorous, evidence-grounded, community-governed foundation — the first unified architecture for AI governance designed for operational reality.

Section 1. The Architecture Imperative

— Why Frameworks Fail Without Integration

The past decade of AI governance activity has produced an unprecedented volume of frameworks, standards, guidelines, and regulatory instruments. The NIST AI Risk Management Framework (2023) introduced a functional decomposition of AI risk across Govern, Map, Measure, and Manage. ISO/IEC 42001:2023 established management system requirements for AI governance aligned to the ISO high-level structure. The EU AI Act (Regulation 2024/1689) created the first comprehensive binding legal framework for AI risk classification in a major jurisdiction. The OECD AI Principles (2019, revised 2024) provided the international normative foundation for trustworthy AI. Together, these instruments represent an extraordinary achievement of the global governance commons.

And yet AI systems continue to fail in ways that should be governed. Behavioral drift goes undetected for months. Supply chain vulnerabilities propagate from third-party model components into production systems. Emergent capabilities appear without warning, outside the scope of risk assessments conducted at design time. Adversarial attacks succeed against systems that were considered well-governed under their applicable frameworks. Regulators in different jurisdictions impose conflicting requirements on the same system. Governance documentation, when it exists, lacks evidentiary integrity and cannot support audit or accountability functions.

These failures are not failures of individual governance controls. They are failures of architectural integration. Each major framework was designed to address a specific governance problem with depth and rigor. None was designed to compose with the others into a unified operational system. The NIST AI RMF does not specify how governance decisions made at design time are enforced at execution time. ISO/IEC 42001 does not address behavioral drift in deployed systems. The EU AI Act does not specify how conformity assessments are maintained across multi-year operational horizons. MITRE ATLAS does not integrate with W3C PROV-DM. No existing framework provides an interoperability layer that allows organizations to implement multiple standards without duplicating governance artifacts or creating compliance conflicts.

This is the architecture problem: the gap between the depth of individual frameworks and the breadth of governance required by a complete AI system. The HHI Standards Architecture is designed to close that gap. It is not a replacement for existing frameworks — it is the integration layer that those frameworks have always required. By defining the relationships between governance domains, specifying the technical interfaces between governance mechanisms, and organizing the full governance space into a coherent three-layer architecture, the HHI Standards Architecture enables organizations to implement complete, interoperable, and auditable AI governance for the first time.

Architectural Thesis

The failures of AI governance are not failures of standards rigor. They are failures of architectural integration. A complete AI governance system requires not just strong individual controls, but a coherent architecture that defines how those controls compose, interoperate, and mutually reinforce one another across the full system lifecycle.

The architecture imperative carries a further implication: governance systems must be designed for operational reality. A governance framework that exists only in documentation is not a governance system — it is a compliance artifact. The HHI Standards Architecture is designed for deployment. Every layer, every pillar, and every integration dependency specified herein corresponds to a concrete governance mechanism that organizations can instrument, monitor, audit, and improve. The governance community has produced sufficient frameworks. What it has lacked is the architectural specification that transforms those frameworks into a functioning governance system.

Section 2. The Three-Layer Architecture

The HHI Standards Architecture is organized into three hierarchical layers, each representing a distinct level of governance function. The layers are not independent: Layer 1 establishes the substrate upon which Layer 2 operates, and Layer 2 provides the operational context within which Layer 3's epistemic and contextual standards are applied. Removing or weakening any layer degrades the integrity of the layers that depend upon it.

LAYER 1 — Foundational Governance Infrastructure

Layer 1 establishes the architectural substrate of the HHI Standards Architecture. The four pillars of this layer define the conditions of possibility for all other governance functions. Without trust ecosystem design, there is no stakeholder basis for governance authority. Without runtime provenance, governance decisions cannot be traced or verified. Without supply chain integrity, the governance perimeter is undefined. Without governance interoperability, no organization operating under multiple frameworks can implement governance coherently. Layer 1 pillars must be established before any Layer 2 or Layer 3 governance mechanisms are deployed.

[HHI-WP-2026-001](#)

Trust Ecosystem Design

Synthesis: HHI-WP-2026-001 established that AI trust is not a property of individual systems but of the ecosystems within which those systems operate. Trust must be architected through defined stakeholder relationships, explicit trust boundaries, accountability assignment, and verifiable trust mechanisms including attestation, certification, and continuous monitoring.

Architectural Role: Trust Ecosystem Design is the foundational governance theory of the Architecture. It defines the multi-stakeholder governance model that all subsequent pillars operationalize. Without a designed trust ecosystem, governance mechanisms lack the institutional authority to be enforced.

Integration Dependencies: Governance Interoperability Architecture (Layer 1); Execution-Time Governance Framework (Layer 2); Sector-Specific Governance Kits (Layer 3). Trust assertions generated by the trust ecosystem are consumed by execution-time enforcement mechanisms and verified through governance observability telemetry.

[HHI-WP-2026-007](#)

Runtime Provenance Architecture

Synthesis: HHI-WP-2026-007 operationalized W3C PROV-DM for AI systems, establishing a machine-readable provenance record for every governance-relevant event in an AI system's lifecycle: training data lineage, model artifact provenance, inference-time decision records, and governance action audit trails. The paper specified provenance capture schemas, retention policies, and verification protocols.

Architectural Role: Runtime Provenance is the evidentiary backbone of the Architecture. Every governance claim — that a system was tested, that a decision was explainable, that a drift event was detected and remediated — is only as good as the provenance record that supports it. Provenance data is the currency of governance accountability.

Integration Dependencies: All Layer 2 operational governance mechanisms generate provenance records consumed by this pillar. Governance Observability Architecture (Layer 2) instruments provenance capture. Citation Integrity Standard (Layer 3) governs the provenance of governance documentation itself.

[HHI-WP-2026-010](#)

Supply Chain Integrity Standard

Synthesis: HHI-WP-2026-010 established the governance framework for AI supply chains, introducing the Machine Learning Bill of Materials (ML-BOM) as the foundational supply chain artifact. The paper specified component provenance requirements, third-party model governance expectations, dependency vulnerability management, and SPDX 2.3 alignment for AI-specific supply chain documentation.

Architectural Role: Supply Chain Integrity defines the governance perimeter. An AI system's governance posture is only as strong as the least-governed component in its supply chain. This pillar establishes the outer boundary of the Architecture's scope and the procedures for governing what enters that boundary.

Integration Dependencies: Runtime Provenance Architecture (Layer 1) for component-level traceability; Adversarial Governance Framework (Layer 2) for supply chain threat modeling; Governance Observability Architecture (Layer 2) for continuous supply chain monitoring signals.

[HHI-WP-2026-009](#)

Governance Interoperability Architecture

Synthesis: HHI-WP-2026-009 developed the technical and conceptual architecture for composable compliance: a model in which governance artifacts produced under one framework are recognized, translated, and consumed by other applicable frameworks without duplication or conflict. The paper introduced governance crosswalk specifications, framework adapter patterns, and the HHI Governance Namespace standard.

Architectural Role: Governance Interoperability is the integration bus of the Architecture. It enables organizations operating under NIST AI RMF, ISO/IEC 42001, and the EU AI Act simultaneously to maintain a single set of governance artifacts rather than parallel, redundant compliance programs. Without interoperability, multi-framework governance collapses under its own administrative weight.

Integration Dependencies: All other pillars across all three layers. Every governance artifact produced by every pillar is subject to the interoperability standards defined here. Standards body liaison activities (Section 7) depend on this pillar for cross-framework translation.

LAYER 2 — Operational Governance Systems

Layer 2 comprises the active governance mechanisms that govern AI systems in production. These are the systems that run continuously: monitoring behavior, enforcing constraints, detecting threats, generating observability signals, and maintaining governance integrity across time. Layer 2 systems are instrumentally dependent on Layer 1 infrastructure and collectively generate the operational evidence that Layer 3 epistemic standards evaluate and document.

[HHI-WP-2026-004](#)

Execution-Time Governance Framework

Synthesis: HHI-WP-2026-004 addressed the governance gap between design-time risk assessment and runtime AI behavior, introducing execution-time guardrails, inference-time policy enforcement, decision logging requirements, and real-time constraint checking mechanisms. The paper established that governance must be operationalized at the moment of inference, not only at the point of deployment approval.

Architectural Role: The Execution-Time Governance Framework is the primary enforcement mechanism of the Architecture. It translates governance policy into operational constraints that are actively enforced during AI system operation. All other operational governance systems interact with or depend upon this enforcement layer.

Integration Dependencies: Trust Ecosystem Design (Layer 1) for policy authority; Runtime Provenance Architecture (Layer 1) for decision logging; Governance Observability Architecture (Layer 2) for enforcement telemetry; Sector-Specific Governance Kits (Layer 3) for domain-specific enforcement rules.

[HHI-WP-2026-005](#)

Behavioral Drift Governance Protocol

Synthesis: HHI-WP-2026-005 defined behavioral drift as the statistically significant departure of an AI system's behavior from its governance baseline, and established the protocol for detecting, classifying, reporting, and remediating drift events. The paper specified drift detection metrics, escalation thresholds, remediation decision trees, and baseline re-establishment procedures.

Architectural Role: The Behavioral Drift Governance Protocol is the temporal integrity mechanism of the Architecture. AI systems do not remain static: they drift through distribution shift, feedback loops, and environmental change. Without continuous drift governance, a system that was compliant at deployment may become non-compliant in production without any detectable triggering event.

Integration Dependencies: Governance Observability Architecture (Layer 2) for drift signal collection; Longitudinal Governance Standard (Layer 2) for drift event historical record; Execution-Time Governance Framework (Layer 2) for automated drift response enforcement.

[HHI-WP-2026-011](#)

Emergent Behavior Governance Protocol

Synthesis: HHI-WP-2026-011 addressed the governance challenge posed by AI capabilities that were not present or anticipated at design time, establishing detection methodologies for emergent behavior, classification taxonomies, containment protocols, and governance escalation procedures for emergent capability events in deployed systems.

Architectural Role: The Emergent Behavior Governance Protocol is the novelty response system of the Architecture. It governs the space beyond the risk assessment horizon — the capabilities and behaviors that existing assessments did not and could not anticipate. Without this pillar, the Architecture has no systematic response to the unknown unknowns of AI system behavior.

Integration Dependencies: Behavioral Drift Governance Protocol (Layer 2) for behavioral anomaly signals; Adversarial Governance Framework (Layer 2) for threat classification of emergent behaviors; Longitudinal Governance Standard (Layer 2) for emergent event historical tracking.

Adversarial Governance Framework

Synthesis: HHI-WP-2026-012 established the governance framework for adversarial threats to AI systems, aligned to the MITRE ATLAS knowledge base. The paper specified adversarial risk assessment requirements, red-teaming governance, attack surface documentation, adversarial incident response protocols, and post-incident governance review procedures.

Architectural Role: The Adversarial Governance Framework is the threat posture management system of the Architecture. It governs the deliberate manipulation of AI systems by adversarial actors and ensures that governance mechanisms themselves are resilient to adversarial circumvention.

Integration Dependencies: Supply Chain Integrity Standard (Layer 1) for supply chain threat vectors; Governance Observability Architecture (Layer 2) for adversarial signal detection; Harmonized Cyber Governance (Layer 3) for cybersecurity integration.

Governance Observability Architecture

Synthesis: HHI-WP-2026-015 operationalized the OpenTelemetry specification for AI governance telemetry, establishing instrumentation standards for governance signal collection, metrics definitions, trace schema for governance event chains, and dashboard specifications for governance visibility at the system, portfolio, and enterprise levels.

Architectural Role: Governance Observability is the sensory system of the Architecture. Without instrumented observability, governance is blind: organizations cannot detect drift, verify enforcement, measure maturity, or generate the audit evidence required for accountability. Observability transforms governance from a documentation exercise into an operational discipline.

Integration Dependencies: All Layer 2 operational systems are both producers and consumers of observability signals. Runtime Provenance Architecture (Layer 1) stores observability records. Longitudinal Governance Standard (Layer 2) aggregates observability data across time horizons.

Longitudinal Governance Standard

Synthesis: HHI-WP-2026-008 established the governance framework for AI systems across multi-year operational horizons, specifying periodic governance review requirements, governance artifact versioning, lifecycle event governance (model updates, retraining,

decommissioning), and the longitudinal accountability record required for regulatory audit functions.

Architectural Role: The Longitudinal Governance Standard is the temporal dimension of the Architecture. It governs not what an AI system is doing now, but what it has done, how its governance posture has changed, and how accountability is maintained across the full operational lifecycle including decommissioning.

Integration Dependencies: Behavioral Drift Governance Protocol (Layer 2) for longitudinal drift record; Governance Observability Architecture (Layer 2) for longitudinal telemetry aggregation; Citation Integrity Standard (Layer 3) for governance record evidentiary standards.

LAYER 3 — Epistemic and Contextual Governance Standards

Layer 3 governs the knowledge and regulatory context within which the Architecture operates. These standards ensure that governance is jurisdictionally coherent — responsive to the regulatory environment in which systems are deployed — and epistemically sound — grounded in evidence, accurately cited, and capable of supporting audit and accountability functions. Layer 3 standards do not govern AI system behavior directly; they govern the governance knowledge base from which all other standards derive their authority.

[HHI-WP-2026-002](#)

Global Fragmentation Response

Synthesis: HHI-WP-2026-002 analyzed the divergence between major AI regulatory regimes — the EU AI Act, the US Executive Order on AI, the UK Pro-Innovation Approach, the China AI Governance Framework, and emerging instruments in India, Brazil, and the Gulf states — and proposed a harmonization strategy enabling organizations to maintain a single governance posture across multiple jurisdictions without regulatory arbitrage or compliance failure.

Architectural Role: Global Fragmentation Response is the regulatory mapping system of the Architecture. It ensures that the Architecture's governance mechanisms satisfy applicable legal requirements in each jurisdiction of operation, and that the interoperability layer (Layer 1) is calibrated to the current fragmentation landscape.

Integration Dependencies: Governance Interoperability Architecture (Layer 1) for cross-jurisdictional framework mapping; Sector-Specific Governance Kits (Layer 3) for jurisdiction-specific sectoral requirements; Standards body liaison activities (Section 7) for regulatory input channels.

[HHI-WP-2026-003](#)

Harmonized Cyber Governance

Synthesis: HHI-WP-2026-003 developed the convergence model for AI governance and cybersecurity, arguing that treating these as separate disciplines creates governance gaps exploitable by adversaries. The paper specified the integration points between NIST CSF 2.0 and NIST AI RMF, established AI-specific cybersecurity controls, and defined the organizational model for unified AI security-governance functions.

Architectural Role: Harmonized Cyber Governance is the security integration standard of the Architecture. It ensures that cybersecurity controls and governance mechanisms are aligned, that security events are treated as governance events, and that the Adversarial Governance Framework operates within a coherent cybersecurity context.

Integration Dependencies: Adversarial Governance Framework (Layer 2) for threat intelligence integration; Governance Observability Architecture (Layer 2) for unified security-governance telemetry; Global Fragmentation Response (Layer 3) for cybersecurity regulatory alignment.

HHI-WP-2026-006

Sector-Specific Governance Kits

Synthesis: HHI-WP-2026-006 delivered customized governance kit specifications for four high-priority sectors: healthcare AI (aligned to FDA AI/ML guidance, HIPAA, and EU MDR), financial services AI (aligned to SR 11-7, EBA guidelines, and MiCA), critical infrastructure AI (aligned to NERC CIP and CISA AI guidance), and public administration AI (aligned to EU AI Act public sector provisions and NIST AI RMF government deployment guidance).

Architectural Role: Sector-Specific Governance Kits are the contextual adaptation layer of the Architecture. The Architecture's core mechanisms are sector-agnostic; the governance kits specify how those mechanisms are configured, parameterized, and extended for the risk profiles and regulatory requirements of specific deployment contexts.

Integration Dependencies: Execution-Time Governance Framework (Layer 2) for sector-specific enforcement rule sets; Longitudinal Governance Standard (Layer 2) for sector-specific audit retention requirements; Global Fragmentation Response (Layer 3) for jurisdiction-sector matrix.

HHI-WP-2026-013

Citation Integrity Standard

Synthesis: HHI-WP-2026-013 established the citation standards for AI governance documentation, specifying requirements for source verification, citation format, version specificity, primary source priority, and the treatment of living documents (standards under

revision, regulatory guidance in flux) in governance artifacts intended for audit or regulatory submission.

Architectural Role: Citation Integrity is the epistemic foundation of the Architecture's knowledge base. Governance standards that cite sources inaccurately, incompletely, or without version specificity cannot be audited against those sources. Citation integrity ensures that every governance claim is traceable to a verifiable, versioned source.

Integration Dependencies: Evidence-Anchored Writing Standard (Layer 3) for integrated documentation methodology; Runtime Provenance Architecture (Layer 1) for citation chain provenance; Longitudinal Governance Standard (Layer 2) for citation record versioning across time.

[HHI-WP-2026-014](#)

Evidence-Anchored Writing Standard

Synthesis: HHI-WP-2026-014 established the methodology for governance artifact composition, specifying that every governance claim must be anchored to an evidentiary basis — empirical data, regulatory text, standards provisions, or peer-reviewed research — and that governance documentation must be structured to support the audit functions it is intended to serve.

Architectural Role: Evidence-Anchored Writing is the composition standard for all governance artifacts produced by the Architecture. It ensures that governance documentation is not merely compliant in form but evidentially sound in substance — capable of withstanding regulatory scrutiny, audit challenge, and evidentiary review.

Integration Dependencies: Citation Integrity Standard (Layer 3) for source standards; Governance Observability Architecture (Layer 2) for empirical evidence generation; Longitudinal Governance Standard (Layer 2) for artifact versioning and archival governance.

Section 3. The HHI Governance Maturity Model

The HHI Governance Maturity Model (HHI-GMM) provides a structured assessment framework for organizations implementing the HHI Standards Architecture. The model defines six levels (0 through 5) representing progressive capability in AI governance, from the absence of systematic

governance through full ecosystem participation. The HHI-GMM is designed for use in self-assessment, independent audit, and regulatory evaluation contexts.

Level 0: Ad Hoc

Description: No systematic AI governance exists. Governance actions, if any, are reactive, undocumented, and not repeatable. Risk assessments are absent or informal. No governance roles are formally assigned.

Capability Requirements: None established. Organization may have individual employees with governance awareness but no institutional governance capability.

Key Governance Artifacts: None. The absence of artifacts is itself a diagnostic indicator.

Assessment Criteria: No documented AI governance policy; no assigned governance roles; no systematic risk assessment process; no audit trail for AI decisions.

Transition Requirements to Level 1: Executive governance policy adoption; AI system inventory; designated governance lead; initiation of Layer 1 pillar gap assessment.

Level 1: Documented

Description: Governance artifacts exist: policies, risk registers, system inventories, and governance role assignments. However, these artifacts are not operationalized. Governance exists in documentation but not in practice.

Capability Requirements: Governance policy authorship capability; basic AI system documentation; designated governance lead with governance mandate.

Key Governance Artifacts: AI governance policy document; AI system inventory; initial risk register; governance role assignments; ML-BOM (initiated).

Assessment Criteria: Documented policies covering at minimum trust ecosystem principles, supply chain inventory, and basic risk classification; evidence of governance role assignments; no requirement for operational deployment of governance mechanisms.

Transition Requirements to Level 2: Layer 1 infrastructure deployment (provenance, interoperability baseline); at least one Layer 2 operational system activated; governance observability instrumentation initiated.

Level 2: Operational

Description: Governance systems are active and monitored. At least one Layer 2 operational governance mechanism is deployed and producing governance telemetry. Governance is no longer only a documentation exercise; it is an operational discipline with measurable outputs.

Capability Requirements: MLOps governance engineering capability; governance observability instrumentation; governance telemetry review process; incident response procedure for governance failures.

Key Governance Artifacts: Governance observability dashboard; operational drift detection reports; execution-time governance logs; provenance records for production AI decisions; supply chain ML-BOM (complete).

Assessment Criteria: Active governance telemetry covering at minimum behavioral monitoring and execution-time logging; documented and tested incident response for governance failures; evidence of at least one governance intervention resulting from operational monitoring.

Transition Requirements to Level 3: Full Layer 1 infrastructure deployment; all applicable Layer 2 systems operational; Layer 3 epistemic standards adopted; cross-layer integration testing initiated.

Level 3: Integrated

Description: All three Architecture layers are implemented and interoperable. Governance mechanisms share data, produce coherent audit trails, and function as a unified system rather than a collection of independent controls. Cross-layer integration has been tested and verified.

Capability Requirements: Full governance function model (Section 6) operational; cross-framework interoperability capability; longitudinal governance program; sector-specific governance kit implementation; evidence-anchored documentation methodology adopted.

Key Governance Artifacts: Integrated governance architecture documentation; cross-layer integration test results; governance interoperability crosswalk; longitudinal governance record; sector-specific compliance matrix; full citation integrity-compliant governance corpus.

Assessment Criteria: Evidence of data flow between all three architecture layers; completed cross-framework interoperability mapping; documented and verified longitudinal governance program; governance documentation assessed as citation-integrity-compliant by independent review.

Transition Requirements to Level 4: Established governance improvement cycle; defined governance metrics with historical baselines; at minimum annual governance maturity self-assessment; initiation of external standards certification process.

Level 4: Optimizing

Description: Continuous governance improvement is underway with measurable outcomes. The organization measures its governance posture quantitatively, tracks improvement over time,

and uses governance telemetry to drive structural improvements to AI systems and governance mechanisms alike.

Capability Requirements: Quantitative governance metrics program; governance improvement backlog with prioritization methodology; external audit capability; regulatory engagement capability; publication-ready governance documentation.

Key Governance Artifacts: Governance metrics dashboard with historical trends; annual governance improvement report; external audit findings and remediation records; external standards certification (ISO/IEC 42001 or equivalent); governance improvement roadmap.

Assessment Criteria: Demonstrated improvement in at minimum three quantitative governance metrics over a twelve-month horizon; completed external audit with findings remediated; evidence of governance improvements driven by observability data rather than compliance requirements alone.

Transition Requirements to Level 5: Active participation in at least one external governance community; contribution of governance artifacts or research to the commons; initiation of standards body liaison activities.

Level 5: Ecosystem Participant

Description: The organization is an active contributor to AI governance trust ecosystems and standards bodies. It not only implements the HHI Standards Architecture but contributes to its evolution, shares governance knowledge with the commons, and exercises governance authority at the ecosystem level through standards participation, public comment, and working group leadership.

Capability Requirements: Standards body participation capability; public governance research capability; ecosystem governance leadership; governance canon contribution process adherence; cross-organizational governance collaboration.

Key Governance Artifacts: Published governance research or standards contributions; standards body working group participation records; public comment submissions to regulatory bodies; HHI Canon amendment proposals; ecosystem trust architecture documentation.

Assessment Criteria: Evidence of active participation in at minimum one standards body working group; at minimum one published governance contribution to the commons; documented ecosystem-level governance relationships; demonstrated alignment of organizational governance posture with ecosystem trust requirements.

Section 4. Standards Alignment Master Map

The following table presents the comprehensive alignment of the HHI Governance Canon to the four primary external standards and regulatory frameworks: NIST AI Risk Management Framework 1.0, ISO/IEC 42001:2023, the EU AI Act (Regulation 2024/1689), and NIST Cybersecurity Framework 2.0. Coverage analysis follows the table.

HHI Canon Document	NIST AI RMF 1.0 (Functions)	ISO/IEC 42001:2023 (Clauses)	EU AI Act (Articles)	NIST CSF 2.0 (Functions)
HHI-WP-2026-001 Trust Ecosystem Design	GOVERN 1.0, 2.0, 6.0; MAP 1.0, 2.0	Clauses 4, 5, 6.1, 9.3	Articles 9, 13, 14, 17	GV.OC, GV.RM, GV.SC
HHI-WP-2026-002 Global Fragmentation Response	GOVERN 6.2; MAP 5.0	Clauses 4.1, 4.2, 9.1	Articles 1–4, 51–55, 77–78	GV.OC, GV.PO
HHI-WP-2026-003 Harmonized Cyber Governance	GOVERN 1.7; MAP 2.0; MANAGE 4.0	Clauses 6.1.2, 8.4, 9.2	Articles 9(5), 15	GV.RM, PR.AA, PR.AT, DE.CM, RS.CO
HHI-WP-2026-004 Execution-Time Governance	MANAGE 1.0, 2.0, 3.0; MEASURE 2.0	Clauses 8.3, 8.4, 8.5, 10.2	Articles 9, 10, 12, 13, 15	PR.PS, DE.CM, RS.MI

HHI-WP-2026-005 Behavioral Drift Governance	MEASURE 2.5, 2.6, 2.7; MANAGE 2.2	Clauses 8.4, 9.1, 10.1	Articles 9(7), 12, 15	DE.CM, DE.AE, RS.AN, RS.MI
HHI-WP-2026-006 Sector-Specific Governance	MAP 3.0, 5.0; GOVERN 4.0	Clauses 4.1, 6.1.2, 8.4, Annex A	Articles 6–9, 40–44, 49–50	GV.OC, GV.PO, ID.IM
HHI-WP-2026-007 Runtime Provenance Architecture	MEASURE 1.0, 2.0; GOVERN 1.3	Clauses 7.5, 8.4, 9.1	Articles 12, 17, 18, 73	ID.AM, PR.DS, DE.CM
HHI-WP-2026-008 Longitudinal Governance	GOVERN 5.0; MANAGE 4.0; MEASURE 4.0	Clauses 9.1, 9.2, 9.3, 10.3	Articles 9(7), 17, 61, 72	GV.OC, ID.IM, RC.CO
HHI-WP-2026-009 Governance Interoperability	GOVERN 6.2; MAP 5.0	Clauses 4.4, 6.2, 8.1	Articles 77–78, Recitals 9–12	GV.PO, GV.SC
HHI-WP-2026-010 Supply Chain Integrity	MAP 1.6, 2.0; GOVERN 6.2; MANAGE 2.4	Clauses 8.3, 8.4, Annex A.5	Articles 10, 13, 25, 28	GV.SC, ID.AM, PR.PS

HHI-WP-2026-01 1 Emergent Behavior Governance	MEASURE 2.5; MAP 1.5; MANAGE 3.2	Clauses 6.1.2, 8.4, 10.1	Articles 9(7), 15, 72	DE.AE, RS.AN, RS.MI
HHI-WP-2026-01 2 Adversarial Governance	MEASURE 2.6; MANAGE 4.0; MAP 2.2	Clauses 6.1.2, 8.4, 9.2	Article 15(1),(2)	PR.AA, DE.CM, DE.AE, RS.MI, RC.RP
HHI-WP-2026-01 3 Citation Integrity Standard	GOVERN 1.3; MEASURE 1.0	Clauses 7.5, 9.1	Articles 17, 73 (documentation)	GV.PO, ID.IM
HHI-WP-2026-01 4 Evidence-Anchored Writing	GOVERN 1.3; MEASURE 1.0, 4.0	Clauses 7.5, 9.1, 9.3	Articles 11, 17, 18	GV.PO, ID.IM
HHI-WP-2026-01 5 Governance Observability	MEASURE 2.0, 3.0, 4.0; GOVERN 5.2	Clauses 8.4, 9.1, 9.2	Articles 12, 15, 72	DE.CM, DE.AE, ID.IM
HHI-WP-2026-01 6 Standards Architecture (This Document)	All functions — architectural integration	All clauses — management system integration	All applicable articles — unified compliance architecture	All functions — unified security-governan ce posture

Coverage Analysis

NIST AI RMF Coverage: The HHI Governance Canon provides comprehensive coverage of all four NIST AI RMF core functions (GOVERN, MAP, MEASURE, MANAGE). The Canon extends beyond the RMF in three domains: runtime provenance instrumentation specifications, adversarial governance aligned to MITRE ATLAS (which post-dates the RMF's primary development), and governance observability operationalization via OpenTelemetry. These extensions represent areas where the HHI Canon offers implementation specificity that the RMF's principles-based design intentionally defers.

ISO/IEC 42001:2023 Coverage: The Canon provides strong coverage of ISO/IEC 42001's management system requirements (Clauses 4–10) and Annex A controls. The Canon extends beyond 42001 in: behavioral drift governance (42001 addresses monitoring but not drift classification and remediation protocols), emergent behavior governance (not addressed in 42001), and longitudinal governance across multi-year operational horizons.

EU AI Act Coverage: The Canon addresses the primary compliance obligations of the EU AI Act for high-risk AI systems, including risk management (Article 9), data governance (Article 10), technical documentation (Article 11), transparency (Articles 13–14), and post-market monitoring (Article 72). The Canon extends beyond the Act's current scope in: supply chain ML-BOM specifications (beyond Article 28 obligations), adversarial governance against MITRE ATLAS (beyond Article 15 robustness requirements), and governance observability instrumentation standards.

NIST CSF 2.0 Coverage: The Canon provides strong coverage across all six CSF 2.0 functions (GOVERN, IDENTIFY, PROTECT, DETECT, RESPOND, RECOVER). HHI-WP-2026-003's harmonized cyber governance specifically bridges CSF 2.0 and the NIST AI RMF, addressing a documented gap in both frameworks' standalone guidance.

Section 5. Implementation Architecture

— Deploying the HHI Standards Architecture

Deploying the HHI Standards Architecture is a multi-year enterprise transformation program. The following four-phase implementation architecture provides the sequencing, capability requirements, governance artifact deliverables, and success criteria for organizations undertaking full Architecture deployment. Phases are designed to overlap: Phase 2 activities begin before Phase 1 is complete, and Phase 3 activities begin before Phase 2 concludes, reflecting the reality that governance capability builds incrementally and continuously.

Phase 1 — Governance Foundation (Months 1–6)

Objective: Establish the Layer 1 foundational infrastructure. By the end of Phase 1, the organization has a functioning trust ecosystem model, a complete AI system and supply chain inventory, an operational provenance architecture, and a governance interoperability baseline.

- **Trust Ecosystem Mapping:** Identify all AI systems in scope; map stakeholders, trust boundaries, and accountability assignments for each system; adopt HHI-WP-2026-001 trust ecosystem design methodology; produce trust architecture documentation for each high-risk system.
- **Supply Chain Inventory and ML-BOM Generation:** Conduct complete inventory of AI components including foundation models, third-party datasets, pre-trained components, and inference infrastructure; generate ML-BOM in SPDX 2.3 format for each system; establish supply chain monitoring procedures.
- **Provenance Architecture Deployment:** Deploy W3C PROV-DM-aligned provenance capture for all governance-relevant AI system events; establish provenance storage and retention infrastructure; verify provenance record integrity for a sample of production AI decisions.
- **Governance Interoperability Baseline:** Conduct gap assessment against applicable frameworks (NIST AI RMF, ISO/IEC 42001, EU AI Act as applicable); produce governance crosswalk mapping; establish HHI Governance Namespace; identify artifact reuse opportunities across frameworks.

Required Organizational Capabilities: Designated CAIGO or interim governance lead; ML system documentation capability; legal/regulatory counsel for framework applicability assessment; MLOps engineering capability for provenance instrumentation.

Governance Artifacts Produced: Trust ecosystem architecture documentation; complete AI system inventory; ML-BOM for each in-scope system; provenance capture deployment documentation; governance interoperability crosswalk; Phase 1 governance maturity assessment (baseline).

Success Criteria: 100% of in-scope AI systems inventoried; ML-BOM generated for all high-risk systems; provenance capture active for at minimum one production system; governance interoperability crosswalk completed for all applicable frameworks; HHI-GMM Level 1 assessment achieved.

Phase 2 — Operational Governance (Months 6–18)

Objective: Deploy Layer 2 operational governance systems. By the end of Phase 2, the organization has active execution-time governance, behavioral drift monitoring, governance observability telemetry, and an adversarial governance program across all high-risk AI systems.

- **Execution-Time Governance Deployment:** Instrument inference-time policy enforcement for all high-risk systems; deploy decision logging to provenance

infrastructure; establish governance constraint violation alerting; document enforcement rule sets by system and sector.

- **Behavioral Drift Monitoring Activation:** Establish governance baselines for all monitored behavioral metrics; deploy drift detection algorithms; configure escalation thresholds and notification procedures; conduct tabletop exercise for drift remediation decision tree.
- **Governance Observability Instrumentation:** Deploy OpenTelemetry-aligned governance telemetry collectors; establish governance metrics definitions and measurement procedures; configure governance observability dashboards at system, portfolio, and enterprise levels; initiate longitudinal telemetry baseline.
- **Adversarial Governance Framework Deployment:** Conduct MITRE ATLAS-aligned adversarial risk assessment for all high-risk systems; establish red-teaming governance program; deploy adversarial incident detection and response procedures; integrate adversarial signals with governance observability.

Required Organizational Capabilities: MLOps governance engineering team; governance observability engineering; adversarial testing capability (internal or contracted); governance incident response team; behavioral analytics capability.

Governance Artifacts Produced: Execution-time governance rule sets by system; behavioral drift baselines and monitoring reports; governance observability dashboard documentation; adversarial risk assessment reports; adversarial incident response playbook; Phase 2 governance maturity assessment.

Success Criteria: Execution-time governance active for all high-risk systems; drift monitoring active with at minimum 30-day baseline established; governance observability telemetry flowing for all monitored systems; adversarial risk assessment completed for all high-risk systems; first governance intervention resulting from operational monitoring documented; HHI-GMM Level 2 assessment achieved.

Phase 3 — Contextual Governance (Months 12–24)

Objective: Implement Layer 3 epistemic and contextual governance standards. By the end of Phase 3, the organization has sector-specific governance kits deployed, a longitudinal governance program established, and governance documentation meeting citation integrity and evidence-anchored writing standards.

- **Sector-Specific Governance Kit Implementation:** Select and configure applicable HHI-WP-2026-006 governance kits for each deployment sector; integrate sector-specific enforcement rules with execution-time governance; produce sector compliance matrices; establish regulatory liaison procedures for each applicable jurisdiction.
- **Longitudinal Governance Program Establishment:** Define governance review cycle (minimum annual for high-risk systems); establish governance artifact versioning and

archival procedures; deploy longitudinal governance record in provenance infrastructure; conduct first periodic governance review under the longitudinal standard.

- **Citation and Evidence Writing Standards Adoption:** Audit existing governance documentation corpus against HHI-WP-2026-013 citation integrity requirements; remediate citation deficiencies; train governance documentation authors in evidence-anchored writing methodology; establish governance documentation review process incorporating citation verification.
- **Fragmentation and Harmonization Strategy:** Assess applicable jurisdictions for each AI system; apply HHI-WP-2026-002 harmonization strategy; produce jurisdiction-specific compliance matrices; establish regulatory change monitoring process.

Required Organizational Capabilities: Sector compliance expertise; regulatory affairs capability; governance documentation authorship team trained in citation integrity and evidence-anchored writing; longitudinal record management capability.

Governance Artifacts Produced: Sector-specific governance kit configuration documentation; sector compliance matrices; longitudinal governance program documentation; governance documentation corpus meeting citation integrity standards; jurisdiction compliance matrices; regulatory change monitoring register.

Success Criteria: Sector governance kits deployed for all applicable sectors; longitudinal governance program initiated with first review cycle completed; governance documentation corpus assessed as citation-integrity-compliant by independent review; jurisdiction compliance matrices completed for all applicable regulatory jurisdictions; HHI-GMM Level 3 assessment achieved.

Phase 4 — Architecture Integration (Months 18–36)

Objective: Achieve full cross-layer integration, governance maturity advancement, external standards certification, and ecosystem participation. By the end of Phase 4, the organization operates the HHI Standards Architecture as a unified, continuously improving governance system.

- **Cross-Layer Integration Testing:** Conduct integration testing verifying data flow and governance coherence across all three architecture layers; identify and remediate integration gaps; produce integration test documentation for audit purposes; establish continuous integration testing cadence.
- **Governance Maturity Assessment:** Conduct formal HHI-GMM assessment (self-assessment with independent verification); produce governance maturity report with quantitative metrics; establish improvement roadmap targeting HHI-GMM Level 4; submit maturity assessment to HHI Canon publication record (optional).
- **External Standards Certification Pursuit:** Initiate ISO/IEC 42001:2023 certification process; conduct NIST AI RMF profile alignment assessment; engage EU AI Act

conformity assessment body as applicable; document certification progress in longitudinal governance record.

- **Ecosystem Participation:** Identify applicable standards body working groups (ISO/IEC JTC 1/SC 42, NIST AI RMF working groups, EU AI Act technical committees); initiate participation; contribute governance artifacts or research to the commons as appropriate; pursue HHI Canon working group participation.

Required Organizational Capabilities: Independent governance audit capability; standards body participation capability; public governance research or documentation capability; executive sponsorship for certification investments.

Governance Artifacts Produced: Cross-layer integration test results; formal HHI-GMM maturity assessment report; external certification documentation; standards body participation records; ecosystem governance architecture documentation; full Architecture deployment documentation suitable for external publication.

Success Criteria: Cross-layer integration verified; HHI-GMM Level 3 achieved (Level 4 targeted); at minimum one external certification initiated; active participation in at minimum one standards body working group; HHI-GMM Level 4 assessment achieved by end of program.

Section 6. Governance Roles and Responsibilities Within the Architecture

The HHI Standards Architecture requires a defined organizational governance function model to operate. The following role specifications define the responsibilities, authorities, required competencies, and key governance artifact ownership for each function in the model.

Organizations may combine roles depending on size and AI system portfolio scope, provided that accountability assignments remain unambiguous and conflicts of interest are managed.

Chief AI Governance Officer (CAIGO)

Responsibilities: Enterprise-level accountability for HHI Standards Architecture deployment; governance policy adoption and maintenance; executive reporting on governance posture; regulatory engagement and standards body participation at the enterprise level; governance investment and resource authorization; final escalation authority for governance incidents.

Authorities: Authority to halt AI system deployment pending governance remediation; authority to commit organizational resources to governance programs; authority to represent the organization in external governance fora.

Required Competencies: Advanced knowledge of AI governance frameworks (NIST AI RMF, ISO/IEC 42001, EU AI Act); executive leadership and organizational change management; regulatory affairs experience; sufficient technical understanding to evaluate MLOps governance engineering outputs.

Key Governance Artifacts Owned: Enterprise AI governance policy; governance maturity assessment report; executive governance dashboard; standards body participation records; regulatory correspondence.

AI System Governance Lead

Responsibilities: Per-system accountability for Architecture implementation; system-level governance artifact production and maintenance; coordination of system governance review cycles; escalation of system-level governance incidents to CAIGO; interface with sector compliance and MLOps engineering for system-specific governance requirements.

Authorities: Authority to initiate system-level governance incident response; authority to require governance artifact updates from system owners; authority to recommend system operational pause to CAIGO pending governance remediation.

Required Competencies: Deep knowledge of applicable governance frameworks for the system's risk classification and sector; ML system lifecycle understanding; governance artifact authorship to citation integrity and evidence-anchored writing standards; risk assessment methodology.

Key Governance Artifacts Owned: System trust ecosystem architecture documentation; system ML-BOM; system risk register; system governance review records; system provenance documentation; sector compliance matrix.

MLOps Governance Engineer

Responsibilities: Technical instrumentation of all Layer 2 operational governance systems; provenance capture deployment and maintenance; execution-time governance rule implementation; governance observability pipeline engineering; drift detection algorithm deployment and configuration; integration testing for cross-layer governance data flows.

Authorities: Authority to implement and modify governance instrumentation in production systems (subject to change management); authority to flag governance telemetry anomalies directly to AI System Governance Lead.

Required Competencies: MLOps engineering proficiency; OpenTelemetry and distributed systems observability; W3C PROV-DM and provenance engineering; execution-time policy enforcement implementation; adversarial ML security engineering; behavioral analytics.

Key Governance Artifacts Owned: Governance observability instrumentation documentation; execution-time enforcement rule set technical specifications; drift detection configuration documentation; provenance capture schema documentation; integration test results.

Governance Audit Lead

Responsibilities: Independent verification of Architecture implementation against HHI standards and applicable external frameworks; governance documentation quality review against citation integrity and evidence-anchored writing standards; preparation of audit findings reports; coordination of external certification audits; management of audit finding remediation tracking.

Authorities: Authority to access all governance artifacts, telemetry, and documentation without restriction; authority to issue audit findings with remediation requirements; independence from operational governance functions is a structural requirement of this role.

Required Competencies: Audit methodology and independence standards; comprehensive knowledge of all applicable governance frameworks; governance documentation review against citation integrity standards; external certification audit experience (ISO/IEC 42001, SOC 2, or equivalent).

Key Governance Artifacts Owned: Internal governance audit reports; external certification audit coordination documentation; audit finding remediation tracking register; governance documentation quality assessment reports.

Sector Compliance Officer

Responsibilities: Sector-specific regulatory requirements analysis and translation into Architecture implementation requirements; sector governance kit configuration and maintenance; regulatory change monitoring for applicable sectors and jurisdictions; liaison with sector regulators and industry working groups; sector compliance matrix maintenance.

Authorities: Authority to require sector-specific governance controls in excess of baseline Architecture requirements where regulatory obligations demand; authority to represent the organization in sector regulatory proceedings.

Required Competencies: Deep expertise in applicable sector regulatory requirements (healthcare: FDA, HIPAA; financial services: SR 11-7, EBA; critical infrastructure: NERC CIP, CISA; public administration: EU AI Act public sector provisions); cross-jurisdictional regulatory awareness; sector industry working group participation.

Key Governance Artifacts Owned: Sector compliance matrices; regulatory liaison correspondence; sector governance kit configuration documentation; regulatory change impact assessments; sector-specific risk assessments.

Section 7. The HHI Governance Canon

— Publication and Contribution Framework

The HHI Governance Canon is a living standards corpus. This section defines the governance of the Canon itself: how it is maintained, how it evolves, how the community contributes, and how it interfaces with the broader standards ecosystem.

7.1 Version Cadence

The Canon follows semantic versioning (MAJOR.MINOR.PATCH). Major versions (e.g., v1.0.0 → v2.0.0) are issued annually following a comprehensive review of each whitepaper against the current state of the AI governance landscape, regulatory developments, and community feedback. Minor versions (e.g., v1.0.0 → v1.1.0) are issued quarterly to address significant standard updates, regulatory changes, or evidence-based corrections. Patch versions address typographical errors and minor citation corrections and may be issued at any time. All versions are deposited to Zenodo with DOI minting and committed to the GitHub repository with version tags.

7.2 Amendment Process

Amendments to Canon documents follow a structured evidence-based process: (1) Amendment Proposal submission via GitHub Pull Request, including a structured rationale citing the evidence basis for the proposed change; (2) Editorial Board review for scope and evidence quality; (3) Public comment period (30 days for minor amendments, 60 days for substantive changes); (4) Working group deliberation and disposition; (5) Editorial Board approval; (6) Version release with amendment notes. All amendment proposals must comply with the Citation Integrity Standard (HHI-WP-2026-013) and Evidence-Anchored Writing Standard (HHI-WP-2026-014).

7.3 Community Contribution

The Canon is community-governed with HHI editorial authority. Community members may contribute through: (1) Public comment during designated comment periods; (2) Working group participation in domain-specific Canon working groups; (3) Amendment proposals via GitHub; (4) Research contributions that inform future Canon documents; (5) Implementation case studies contributed to the HHI knowledge base. All contributors are acknowledged in the Canon's contributor registry and credited in relevant whitepaper acknowledgment sections.

7.4 Standards Body Liaison

HHI maintains active liaison with the following standards bodies and working groups: ISO/IEC JTC 1/SC 42 (Artificial Intelligence); NIST AI RMF Working Groups; EU AI Act Technical Committees and Notified Bodies; IEEE Standards Association P7000-series working groups; W3C Data Provenance and AI standards activities; OECD.AI Policy Observatory. Liaison activities are documented and reported in the annual HHI Standards Liaison Report, a public document deposited to Zenodo and published on the HHI GitHub repository.

7.5 Zenodo DOI Management

Each whitepaper and each major Canon release receives a persistent Zenodo DOI. Version history is maintained through Zenodo's versioning infrastructure, ensuring that citations to specific Canon versions remain resolvable permanently. The Zenodo collection record for the HHI Governance Canon aggregates all sixteen whitepapers and the annual integrated Canon release under a single collection DOI, facilitating citation of the Canon as a unified corpus.

7.6 GitHub Repository Governance

The HHI GitHub repository (github.com/HollowHouseInstitute/governance-canon) operates under the following governance rules: main branch is protected with required editorial board review for all merges; all substantive changes require at minimum two reviewer approvals; release tags correspond to Zenodo deposits; issue tracking is used for all amendment proposals and community feedback; branch naming conventions and commit message standards are specified in the repository CONTRIBUTING.md; all repository changes are logged in the CHANGELOG.md with version attribution.

Section 8. The Future of the HHI Governance Canon

The HHI Governance Canon v1.0.0 addresses the AI governance landscape as it exists in 2026. The frontier of AI development is advancing faster than governance capacity, and several emerging domains require Canon expansion before adequate governance architecture can be claimed. The following five research frontiers constitute the HHI Canon v2.0 research agenda.

8.1 Agentic AI Governance

Problem Statement: Multi-agent autonomous AI systems — systems in which multiple AI models collaborate, delegate, and execute actions without continuous human oversight — present governance challenges that the current Canon does not fully address. Agentic systems exhibit emergent behavior at the system-of-systems level, create novel accountability

assignment problems when multiple agents contribute to a harmful outcome, and may operate across trust boundaries in ways that the current trust ecosystem design framework does not govern.

HHI Research Agenda: Develop an Agentic AI Governance Standard addressing multi-agent trust architecture, inter-agent accountability assignment, agentic behavior observability, and the governance implications of AI-to-AI communication channels that bypass human oversight mechanisms.

8.2 Quantum Computing Governance

Problem Statement: Quantum-accelerated AI introduces governance implications in three domains: the acceleration of adversarial capabilities against AI governance mechanisms (particularly cryptographic protections for provenance and audit records); the potential for quantum AI systems to exhibit behavioral characteristics fundamentally different from classical neural network architectures; and the supply chain integrity implications of quantum computing components in AI infrastructure.

HHI Research Agenda: Develop a Quantum-AI Governance Standard addressing quantum-resistant provenance cryptography, adversarial governance for quantum-accelerated threats, and the governance framework for quantum-classical hybrid AI systems.

8.3 Neuromorphic and Novel Architecture Governance

Problem Statement: Neuromorphic computing architectures, spiking neural networks, and other novel AI hardware and software architectures may exhibit behavioral characteristics — including learning behavior in deployment — that the current Canon's behavioral drift and emergent behavior governance protocols were not designed to address. The measurement and observability infrastructure specified in the Canon assumes classical von Neumann computing substrates.

HHI Research Agenda: Develop a Novel Architecture Governance Standard specifying governance requirements for AI systems operating on neuromorphic, photonic, analog, and other non-classical computing substrates, with particular attention to observability, provenance, and behavioral baseline establishment.

8.4 AI Governance in Space and Extraterrestrial Applications

Problem Statement: AI systems operating in space environments present unique governance challenges: communication latency precludes real-time human oversight; the extreme environment makes behavioral drift governance measurement difficult; jurisdictional frameworks do not clearly apply beyond national airspace; and the consequences of AI governance failures in space systems may be irreversible. The Outer Space Treaty and related instruments do not address AI governance.

HHI Research Agenda: Develop a Space AI Governance Standard addressing autonomous operation governance under communication latency constraints, jurisdictional governance for AI

systems in orbital and extraterrestrial environments, and the governance framework for AI systems with irreversible failure modes.

8.5 Synthetic Biology and AI Convergence Governance

Problem Statement: The convergence of AI systems with synthetic biology — including AI-designed genetic sequences, AI-driven protein structure prediction applied to bioengineering, and AI systems integrated with biological computing substrates — creates governance domains that neither the AI governance community nor the biosafety community has fully addressed. The dual-use potential of AI-synthetic biology convergence is among the most serious governance gaps in the current landscape.

HHI Research Agenda: Develop a Biosynthetic AI Convergence Governance Standard in collaboration with biosafety and biosecurity governance bodies, addressing dual-use risk governance, responsible disclosure protocols, and the integration of AI governance with existing biosafety regulatory frameworks.

Section 9. Recommendations

The following recommendations are addressed to the enterprise AI governance community, regulatory bodies, standards organizations, and the global AI governance commons. They represent the policy-forward conclusions of the HHI Governance Canon, grounded in the evidence developed across the sixteen-paper series.

1. **Adopt the HHI Standards Architecture as an integration framework.** Enterprises operating under multiple AI governance frameworks (NIST AI RMF, ISO/IEC 42001, EU AI Act) should adopt the HHI Governance Interoperability Architecture (HHI-WP-2026-009) to eliminate redundant compliance efforts and produce a unified, auditable governance artifact corpus. Multi-framework governance without an interoperability architecture is not sustainable at enterprise scale.
2. **Mandate runtime provenance for all high-risk AI systems.** Regulators should require machine-readable W3C PROV-DM-aligned provenance records for all AI systems classified as high-risk under applicable frameworks. Governance accountability without provenance is unverifiable. The EU AI Act's technical documentation requirements (Article 17) should be extended with specific provenance instrumentation standards aligned to HHI-WP-2026-007.
3. **Establish behavioral drift governance as a post-market monitoring requirement.** The AI governance community must recognize that a system compliant at deployment may become non-compliant in production through behavioral drift without any discrete triggering event. Post-market monitoring requirements under the EU AI Act (Article 72) and NIST AI RMF's Manage function should be supplemented with specific behavioral drift governance protocols aligned to HHI-WP-2026-005, including drift metric definitions, escalation thresholds, and remediation requirements.

4. **Integrate ML supply chain governance into all AI risk management frameworks.** Supply chain integrity is not an optional governance extension — it is a foundational requirement. Standards bodies updating NIST AI RMF, ISO/IEC 42001, and related frameworks should incorporate ML-BOM requirements, third-party component governance obligations, and SPDX 2.3 alignment as core standard requirements, not informative guidance.
5. **Adopt governance observability as an operational standard for AI systems in production.** The absence of governance observability infrastructure means that governance functions cannot verify their own effectiveness. Enterprises should adopt OpenTelemetry-aligned governance telemetry as specified in HHI-WP-2026-015, and regulators should consider governance observability infrastructure a prerequisite for high-risk AI system deployment authorization.
6. **Develop and enforce adversarial governance requirements aligned to MITRE ATLAS.** The MITRE ATLAS framework provides the most comprehensive taxonomy of adversarial ML tactics and techniques currently available. Governance frameworks should require ATLAS-aligned adversarial risk assessments for high-risk AI systems, red-teaming governance programs, and adversarial incident response plans. Cybersecurity and AI governance must be treated as a unified domain as specified in HHI-WP-2026-003.
7. **Invest in governance harmonization infrastructure to address regulatory fragmentation.** The current trajectory of AI regulatory divergence between the EU, US, UK, China, and other jurisdictions creates unsustainable compliance burdens for globally operating organizations and creates incentives for regulatory arbitrage. International bodies including the OECD, G7, and G20 should prioritize AI governance harmonization efforts, using the HHI Global Fragmentation Response (HHI-WP-2026-002) harmonization strategy as an input to multilateral governance alignment efforts.
8. **Establish the CAIGO as a standard enterprise governance role.** The Chief AI Governance Officer role defined in Section 6 of this document should be recognized as a standard enterprise function with defined authority, accountability, and organizational positioning. Regulatory guidance and corporate governance standards should specify CAIGO-equivalent accountability requirements for organizations operating high-risk AI systems, with accountability chains that reach to board level.
9. **Require citation integrity and evidence-anchored documentation for regulatory submissions.** AI governance documentation submitted to regulatory bodies, notified bodies, or audit functions must meet citation integrity standards (HHI-WP-2026-013) and evidence-anchored writing requirements (HHI-WP-2026-014). Regulatory bodies should publish documentation quality standards aligned to these HHI standards to reduce the volume of non-auditable governance documentation currently submitted in regulatory proceedings.
10. **Fund and prioritize governance research for emerging AI domains.** The five frontier research areas identified in Section 8 — agentic AI, quantum-AI convergence, neuromorphic governance, space AI, and biosynthetic AI convergence — require dedicated research investment now, before deployment at scale. Governments, foundations, and research institutions should prioritize pre-competitive governance

research in these domains, and standards bodies should initiate working groups to address governance gaps before they become governance crises.

11. **Implement longitudinal governance programs for all AI systems with multi-year operational horizons.** AI systems do not remain static, and governance programs that do not account for the evolution of a system over time are fundamentally incomplete. Every organization operating AI systems for more than twelve months should implement a longitudinal governance program aligned to HHI-WP-2026-008, including periodic governance reviews, lifecycle event governance, and a longitudinal accountability record suitable for regulatory audit.
 12. **Contribute to the HHI Governance Canon as a community-governed standard.** The Canon's value as a governance commons depends on the breadth and quality of community contribution. Organizations that have implemented the Architecture should share implementation case studies, amendment proposals, and research contributions through the HHI GitHub repository and working group participation. The standards community should treat the Canon as a contribution to the global governance commons, not as a proprietary product of a single institution.
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Conclusion

The HHI Governance Canon began with a question that the global AI governance community had not answered with sufficient architectural precision: what does a complete AI governance system actually look like? The sixteen whitepapers of the Canon, from the Trust Ecosystem Design of HHI-WP-2026-001 to the Governance Observability Architecture of HHI-WP-2026-015, have each contributed a foundational answer to a specific dimension of that question. This capstone document has assembled those answers into an architecture.

The HHI Standards Architecture is not the only possible answer to the architecture question. It is, to the best of the Institute's knowledge, the first comprehensive and operationally deployable one. It is built from first principles, grounded in the evidence developed across the Canon's research program, calibrated against the major existing governance frameworks, and designed for the organizations that must actually govern AI systems in production: not in theory, not in a research environment, but in the complex, adversarial, regulatorily fragmented, organizationally contested reality of enterprise AI deployment in 2026.

Amy founded the Hollow House Institute on the conviction that governance is not a compliance exercise — it is an architectural discipline. That conviction is the intellectual foundation of every whitepaper in this Canon. It is expressed most fully here, in this synthesis: the recognition that individual governance controls, however rigorously designed, cannot substitute for architectural thinking about how those controls compose, interoperate, and mutually reinforce. The Architecture is Amy's primary contribution to the governance commons, and the Canon is the evidence base that supports it.

The Canon is not finished. The five research frontiers identified in Section 8 will require Canon expansion before adequate governance of the next generation of AI systems can be claimed. The Canon's version cadence, amendment process, and community contribution framework ensure that it will evolve in response to that frontier, guided by the same evidence-based, citation-integrity-governed methodology that produced it. The Canon is released under CC BY 4.0: it belongs to the community that will extend it.

The HHI Governance Canon represents the first unified standards architecture for AI governance. It is an open invitation: to the enterprise that needs to implement it, to the regulator that needs to reference it, to the researcher that needs to critique it, and to the governance professional that needs to improve it. The architecture is sound. The work continues. — Hollow House Institute, Arlington, TX, May 2026

The Hollow House Institute calls upon enterprises, regulators, standards bodies, and the global AI governance community to engage with the HHI Standards Architecture — to implement it, test it against operational reality, and contribute to its evolution. AI governance is not a problem that any single institution can solve. It is a collective architectural challenge. The Canon is HHI's contribution to that collective effort. The next contribution is yours.

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github.com/HollowHouseInstitute/governance-canon

References

1. National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)* (NIST AI 100-1). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.AI.100-1>
2. International Organization for Standardization & International Electrotechnical Commission. (2023). *ISO/IEC 42001:2023 — Information technology — Artificial intelligence — Management system*. ISO.
3. European Parliament and Council of the European Union. (2024). *Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. Official Journal of the European Union, L 2024/1689. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401689
4. National Institute of Standards and Technology. (2024). *The NIST Cybersecurity Framework 2.0* (NIST CSF 2.0). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.CSWP.29>
5. International Organization for Standardization & International Electrotechnical Commission. (2022). *ISO/IEC 27001:2022 — Information security, cybersecurity and privacy protection — Information security management systems — Requirements*. ISO.
6. IEEE Standards Association. (2021). *IEEE 7000-2021 — IEEE Standard Model Process for Addressing Ethical Concerns during System Design*. IEEE. <https://doi.org/10.1109/IEEESTD.2021.9536679>
7. IEEE Standards Association. (2019). *IEEE 7001-2021 — IEEE Standard for Transparency of Autonomous Systems*. IEEE. <https://doi.org/10.1109/IEEESTD.2021.9445851>
8. Organisation for Economic Co-operation and Development. (2024). *OECD AI Principles* (revised). OECD Publishing. <https://oecd.ai/en/ai-principles>
9. UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*. United Nations Educational, Scientific and Cultural Organization. <https://unesdoc.unesco.org/ark:/48223/pf0000381137>
10. World Wide Web Consortium (W3C). (2013). *PROV-DM: The PROV Data Model* (W3C Recommendation). W3C. <https://www.w3.org/TR/prov-dm/>
11. Linux Foundation. (2022). *SPDX Specification 2.3 — Software Package Data Exchange*. The Linux Foundation. <https://spdx.github.io/spdx-spec/v2.3/>
12. MITRE Corporation. (2023). *MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems)*. MITRE. <https://atlas.mitre.org/>

13. Cloud Native Computing Foundation. (2024). *OpenTelemetry Specification v1.x*. CNCF. <https://opentelemetry.io/docs/specs/otel/>
14. Dafoe, A. (2018). AI governance: A research agenda. *Future of Humanity Institute, University of Oxford*. <https://www.fhi.ox.ac.uk/wp-content/uploads/GovAI-Agenda.pdf>
15. Cihon, P., Maas, M. M., & Floridi, L. (2020). Should artificial intelligence governance be centralised? Design lessons from history. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 228–234. <https://doi.org/10.1145/3375627.3375857>
16. Floridi, L., Cows, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., ... Vayena, E. (2018). An ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and Machines*, 28(4), 689–707. <https://doi.org/10.1007/s11023-018-9482-5>
17. Kearns, M., & Roth, A. (2019). *The Ethical Algorithm: The Science of Socially Aware Algorithm Design*. Oxford University Press.
18. Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., ... Gebru, T. (2019). Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229. <https://doi.org/10.1145/3287560.3287596>
19. Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12), 86–92. <https://doi.org/10.1145/3458723>
20. Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1(9), 389–399. <https://doi.org/10.1038/s42256-019-0088-2>
21. Federal Reserve Board & Office of the Comptroller of the Currency. (2011). *SR 11-7: Supervisory Guidance on Model Risk Management*. Board of Governors of the Federal Reserve System.
22. European Banking Authority. (2022). *EBA Guidelines on the use of remote customer onboarding solutions* (EBA/GL/2022/15). EBA. <https://www.eba.europa.eu/>
23. Papernot, N., McDaniel, P., Goodfellow, I., Jha, S., Celik, Z. B., & Swami, A. (2018). Practical black-box attacks against machine learning. *Proceedings of the ACM Asia Conference on Computer and Communications Security*, 506–519. <https://doi.org/10.1145/3052973.3053009>
24. Widder, D. G., West, S., & Whittaker, M. (2023). Open (for business): Big tech, concentrated power, and the political economy of open AI. *SSRN*. <https://doi.org/10.2139/ssrn.4543807>
25. Hollow House Institute. (2026). *HHI Governance Canon, HHI-WP-2026-001 through HHI-WP-2026-015*. Hollow House Institute. github.com/HollowHouseInstitute/governance-canon